

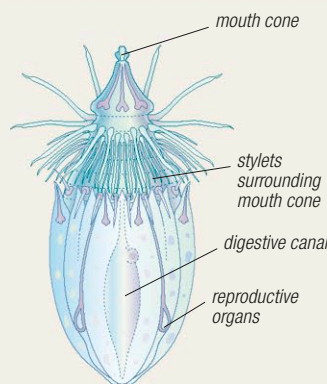
MICRO-ANIMALS

A large part of the animal kingdom consists of animals that are too small to be seen with the naked eye. Some kinds, such as the tardigrades and rotifers, are almost always microscopic, while others have close relatives that are much easier to see. Whatever their size, most have a clearly recognizable body plan, which can be used in their classification. However, occasionally, a micro-animal appears that does not fit any recognized group. One example is the phylum Loricifera, which was added in the late 20th century.



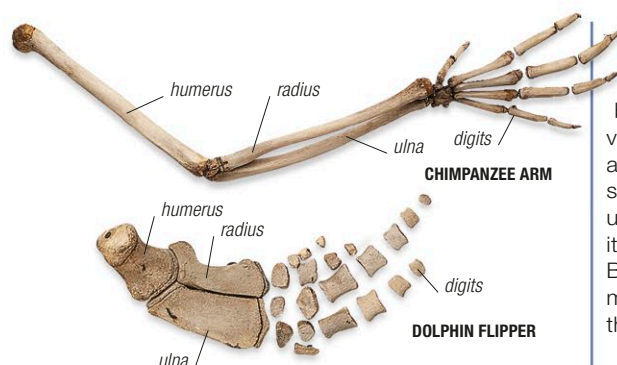
SAMPLING A STREAM

Aquatic habitats often teem with tiny animals, which can be trapped with ultrafine nets. Many are straightforward to classify, but larval forms can cause problems—in the past, they have sometimes been classified as species in their own right.



A NEW FORM OF LIFE

Loriciferans were first described in 1983. Measuring about $\frac{1}{25}$ in (1 mm) long, they live in marine sediments, and have a mouth surrounded by sharp stylets, or spines. They probably live worldwide.



ARMS AND FLIPPERS

A human arm and a dolphin's flipper look very different in life, but they contain the same arrangement of bones. This similarity is strong evidence that chimpanzees (and humans) and dolphins evolved from a common ancestor.

over 300 million years, is built around 3 main sets of bones: a single bone near the body, 2 bones farther out, and 5 sets of smaller bones at the limb's farthest point. As 4-legged vertebrates spread from the water to the land and air, their limbs became very different in shape and size, and in the way they worked, however, the underlying pattern of bones was retained, making it an identifying feature that remains to this day. Because this pattern is unlikely to have evolved more than once, it provides important evidence that these animals evolved from a shared ancestor.

Conflicting clues

Taxonomy is rarely straightforward, because evolution provides a mix of useful and confusing clues. Useful ones include the underlying patterns of bones or body segments, and microscopic

SUPERFICIALLY SIMILAR

Bats and birds both have similar adaptations for flight, such as wings and a lightweight skeleton. However, they are not directly related; bats are mammals and birds have a separate ancestry that originates from among the dinosaurs.



features, such as the way a single cell divides when a new animal starts to take shape. They also include vestigial organs that have lost their original function over time, and become smaller as a result. Among them are the hind limb bones of some snakes that play no part in movement, but indicate a 4-legged ancestry.

Similarity does not always mean genetic relationship. Convergence (p.16) happens when unrelated species evolve similar adaptations to similar ways of life—such as the wings of flying bats and birds. Convergence can sometimes obscure genetic relationships—especially within families or genera. But when examined in more detail, fundamental biological differences can reveal the true pattern of the evolutionary tree and how species have evolved from separate branches.

In extreme cases, convergence can make the relationships between organisms extremely difficult to unravel. A record of the past is built into every living thing in the form of DNA. Since the 1990s, technological advances in both computing and DNA extraction have helped explain convergence and many other questions. By studying the similarities and differences in genetic material, relationships between different organisms can be established regardless of their appearance.

NEW SPECIES

Every year, the world's taxonomists examine thousands of new species that have been collected or photographed in many different parts of the world. Most of them are invertebrates, particularly from the sea and the rainforests, but there are also reptiles, birds, and mammals, often from remote areas of the tropics. Each new species is formally identified, named, and described as part of the classification process.

New discoveries

Since classification first began, taxonomists have identified and catalogued species from all over the world. Today, that process continues. Most large, land-dwelling animals are known, but huge numbers of invertebrates, particularly in the deep ocean, have yet to be discovered and described. Victorian zoologists—including



MICRO LEAF CHAMELEON

Discovered on a tiny islet of Madagascar—and described in 2012—this tiny leaf chameleon ranks among the smallest of land-living vertebrates. Its total length is less than $\frac{1}{4}$ in (3 cm).

famous ones such as Charles Darwin (1809–1882)—thought little about trapping new species, or shooting them out of the trees. In today's more enlightened times, technology plays a role instead. Automatic cameras photograph animals on rain forest floors, and deep in the oceans where there is no natural light. Camera traps helped confirm that the Annamite striped rabbit, on the Laos-Vietnam border, was a genuinely new species.

Splitting species

New species can also be found in another way—by reexamining ones that are already known. Sometimes, this shows that 2 species are actually one, but it can also have the opposite effect. Two species may look similar, but their anatomy, behavior, and genes may show that this similarity is just skin-deep. When this happens, the species is split into 2, and one of them renamed.



OLINGUITO

This cloud forest member of the raccoon family from the Andes was recognized as a separate species after scrutiny of museum skins—one of very few new discoveries of mammalian carnivores.

There are many examples of new species being “discovered” in this way. For example, the African forest elephant (p.109) is now recognized as being different from the African savanna elephant (p.109), whereas formerly they were thought to be one. The giraffe (p.229) was long thought to be a single species, but has now been split into 4 separate species. “Splitting” is particularly common among forest primates, accounting for a considerable growth in the number of species, even though their numbers are often in serious decline.